

A new formulation and an exact approach for the many-to-many hub location-routing problem



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ABSTRACT

An important problem of the freight industry is the parcel delivery network design, where several facilities are responsible for assembling flows from several origins, re-routing them to other facilities where the flows are disassembled and the packages delivered to their final destinations. In order to provide this service, local tours are established for the vehicles assigned to each of the processing facilities, which are then responsible for the pickup and delivery tasks. This application gives rise to the many-to-many hub location routing problem that is the combination of two well known problems: the vehicle routing problem and the single assignment hub location problem. In this work, a new formulation for this important problem is proposed and solved by a specially tailored Benders decomposition algorithm. The proposed method is robust enough to solve instances up to 100 nodes having 4 million integer variables.

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1. Introduction

Freight transportation plays a major role in the economy of any country. It is an important activity that assists industries and trade businesses by transporting and delivering on time raw materials and finished goods. It has therefore a great influence on the overall performance of manufacturing companies as well as on the final costs of products [1]. Because of this importance, freight transportation providers are constantly pressured to lower their costs, while increasing their service levels by meeting tighter schedules, not to mention having to improve driver's wellness and to face rapidly changing economic conditions.

In order to overcome these pressures, freight transportation providers have to regularly redesign and improve their logistic processes by better coordinating the decisions on the strategic, tactical, and operational levels [2]. The coordination of these decisions requires distinct approaches depending on which shipping segment of the freight industry the carrier is competing in.

The freight industry has basically two types of service segments: full truckload (TL) and less-than-truckload (LTL) shipping. While TL is more suitable for customers with large shipment sizes that require individual tailored services such as direct transportation between two points, for instance, between suppliers and manufacturers or manufacturing facilities and warehouses; LTL is more appropriate for clients that cannot be individually serviced by full truckloads and have then to share the carrier's transportation resources with other customers. Usually, for the LTL case, freight transportation providers offer

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regular service schedules that allow freight from different clients, normally located in different origins, to be picked up and then sent to terminal facilities to be sorted and consolidated. The consolidated cargo is then moved towards their destinations through possibly a network of terminals. At the last terminal, the cargo is deconsolidated into smaller components and loaded into vehicles to be finally delivered. For a better description of freight transportation please refer to [3].

One important application in the LTL segment is the transportation of parcels or public postal services [4,5]. Generally, in this application, flow exchanging regional post offices are scattered in a large geographic area. Instead of directly connecting them, which is usually very expensive, local pickup tours, starting and ending at a nearby terminal base, are used to collect the parcels. At the terminal bases, the parcels are sorted, grouped according to their destinations and sent, via more efficient and higher volume carriers, to other terminals where they are deconsolidated and redistributed in order to be delivered to the destinations by means of local delivery tours. While the terminal bases are known as hub facilities or just hubs, the regional post offices are named non-hub nodes. Further, the bulk transportation of large volumes of parcels between hub facilities allows economies of scale to be achieved thus lowering shipping costs. Moreover, the pickup and delivery tours can be planned to happen at the same time by the same vehicle or not depending on the available resources.

The aforementioned description is also the characterization of a specialized version of a hub-and-spoke (HS) structure. Over the last decade, HS networks have attracted the attention of many researchers because of their successful use in cargo and passengers transportation and telecommunication systems [6–11], and more specifically in postal applications [12–15,4,5,16–18].

Originally [19], HS networks were assumed to have a direct connection between every hub pair; no two non-hub nodes could be serviced by a direct link; non-hub nodes could be directly connected to only one of the installed hubs, i.e. were single allocated, or could send and receive flow through more than one of the installed hubs, i.e. were multiple allocated; an origin–destination demand was routed through one or at most two hubs; and economies of scale on inter-hub connections were represented by a discount factor.

However, over the years, these assumptions have been adjusted and adapted to cope with different problems and applications such as: the design of a ring star hub network [20]; the location of inter-hub connections instead of just hubs [21,22]; the minimization of the maximum travel time between any origin–destination pair [23]; allowing at most three hubs on any given route between pairs of origin–destination [24] or incomplete inter-hub connections [25]; but also minimizing the maximum travel time between any origin–destination pair [26]; projecting star-star like hub networks [27]; having a tree structure connecting all installed hubs [28]; and partitioning the points into sub-networks, locating at least one hub in each sub-network and routing the flow at minimum cost [29].

For postal and parcel transportation applications, HS networks have shown to be effective in reducing mail delivery times [18] and lowering operational and infrastructure costs [13,30]. These achievements have been very important since in many countries the deregulation of the sector has increased the competition with third-party companies and express shipment providers [4,18], forcing the postal and parcel transportation providers to redesign their networks in order to lower costs and improve the service level offered.

The deployment of HS ideas to design postal or parcel transportation networks consists of locating the hubs, generating the local tours to service the non-hub nodes and to connect them to the installed hubs, and routing the flow of many origins to many destinations at a minimal cost. The coordination of all of these decisions as one is known as the many-to-many hub location-routing problem (MMHLRP). An example of one solution for the MMHLRP is illustrated in Fig. 1. While the bold triangle and dots are the hubs and non-hubs, the bold and simple edges are the inter-hub connections and local tours, respectively.

Besides being a HS problem and a special area of freight transportation, the MMHLRP is closely related to location-routing problems (LRP), which were initiated by Watson-Gandy and Dohrn [31]. The LRP are actually a set of problems within

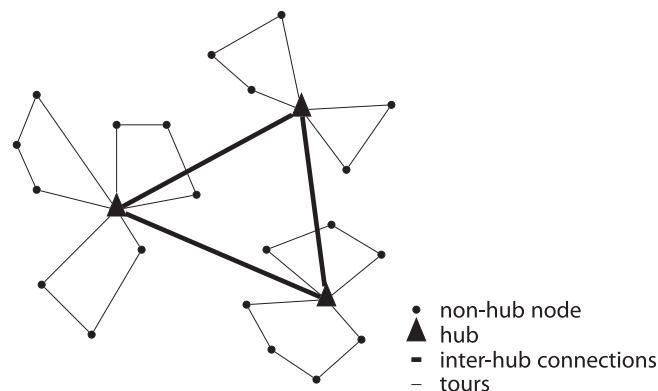


Fig. 1. A MMHLRP solution example.

location analysis with the distinguishing feature of modeling and solving locational problems while addressing issues of vehicle routing or tour planning aspects. Most of these problems consist of locating facilities to service a number of customers by means of vehicle tours. Starting out from a facility, the vehicles visit customers for delivering and/or collecting goods, before returning to the same facility [32].

Generally, while in LRP the facilities are not connected to each other and clients do not exchange flow, in the MMHLRP, the exchange of flows between pairs of clients and the interaction between hubs are two of the main characteristics of the problems. Notwithstanding these differences, in both types of problems, a homogeneous fleet is usually assumed, and the number of available vehicles to deploy and of facilities to be installed are not known a priori [33].

The MMHLRP was first introduced by Nagy and Salhi [15] who proposed a formulation and a hierarchical heuristic to tackle the problem. On one hand, Nagy and Salhi [15] allowed the vehicles to do the pickup and the delivery at the clients on separate occasions, while observing capacity and maximum distance constraints. On the other hand, they disregarded the vehicles' costs and one very important aspect of HS networks: the scale economies in the inter-hub connections. Further, the number of constraints of the presented model grows exponentially with the number of clients, being hard to manage even for small values. Furthermore, although their heuristic has an interesting strategy of locate first and route later embedded in a neighborhood search, the reported results are very limited. Only one instance is solved.

Addressing the restructuring of the Swiss postal service, Bruns et al. [4] used a location model to decide the number, the capacity, and the location of hubs. Bruns et al. [4] did not explicitly consider the routing decisions, but instead they embedded estimations of the routing cost into the formulation.

Wasner and Zäpfel [17] considered the redesign of the Austria postal service. They considered different aspects in their non-linear model, such as the possibility of direct connection between pairs of non-hub nodes, the pickup and delivery happening at the same time, the number of carriers to be used in inter-hub connections, and the presence of a central hub, through which every inter-hub flow must be routed. As the proposed non-linear mixed integer formulation is very hard to manage and solve by standard solvers, they developed a sequence of local search procedures with feedback loops to determine the locations of hubs, the number of transporters, and the local routes. Once again, only one instance is solved.

More recently, Çetiner et al. [18] developed an iterative two-stage solution procedure for locating hubs, planning local tours, and assigning vehicles to the hubs. In their assumptions, Çetiner et al. [18] allowed each non-hub node to be serviced by more than one hub, i.e. to be multiple allocated. They also considered the pickup and the delivery to happen at the same time, while observing only the maximum tour length constraint. In the first stage of their procedure, hub locations are determined and non-hub nodes are multiply allocated to the hubs. In the second stage, multiple traveling salesman problems are solved for each installed hub. The procedure then iterates between the two stages by an updating scheme of the distances used in the first stage. As the solution of the second stage occurs separately for each installed hub, and further as the pickup and the delivery are assumed to happen at the same time, their method produces solutions in which non-hub nodes are visited more than once unnecessarily. They applied their procedure to 7 different test instances ranging from 21 to 49 nodes, and a case study for the Turkish postal service, having 81 nodes.

In this work, we focus on a MMHLRP where there are installation costs for establishing a hub and a vehicle at a hub; there is also a fixed charge for assigning non-hub nodes to hubs, as suggested by Bruns et al. [4] and Cetiner et al. [18] to represent handling costs at the hubs for each customer, implying that non-hub nodes can be serviced by only one hub (single allocated); scale economies are accounted for in inter-hub connections, i.e. a factor $0 < \alpha < 1$ is used to reduce costs between hubs due to bulk transportation; load capacity constraints are disregarded [18], but the service level or the well-being of drivers are respected by observing a maximum time for completing the tours; and the pickup and the delivery are done at the same time by the same vehicle, in other words, each client is visited only once. Further, a homogeneous fleet is assumed.

Our main contributions are the proposition of a manageable but large formulation and an exact algorithm capable of tackling large-scale instances of the classical Australian Post data set having up to 100 customers and 10.000 commodities. The new formulation was devised in a way that allowed the problem to be decomposed into two smaller subproblems. This enabled the efficient deployment of a Benders decomposition algorithm [34] with one keen feature: the concurrent generation of optimality and feasibility cuts. As far as we know, it is the first time that large-scale instances are solved to optimality for such an important problem. Furthermore, the proposed formulation can now be used by practitioners to optimally solve instances up to 30 clients by standard solvers, being also easy to be adapted to different problem settings.

The reminder of the work is organized as follows. Section 2 formally defines the problem, presenting the new formulation and some definitions. The Benders decomposition algorithm is given in Section 3, while the respective experimental results are reported in Section 4. Conclusions follow in Section 5.

2. Formulation and definitions

The new model is based on the combination of two well known formulations in their respective fields of research: One is the model of [35] for the single allocation hub location problem; and the other is the mathematical program proposed by Claus [36] for the traveling salesman problem. They were selected because of the quality of their linear relaxation bounds.

For other formulations for hub location problems and traveling salesman problem, please refer to [11,37], respectively. In order to merge the selected formulations, the following definitions are required.

Let N be the set of client node locations which exchange flows and let K be the set of candidate nodes to become hubs. Usually $K \subseteq N$, but we will assume that $K \equiv N$ henceforth, i.e. all demand nodes are candidates to be a hub. For all node pairs i and j ($i, j \in N : i \neq j$), w_{ij} represents the flow demand from client i to customer j which is routed through either one or two installed hubs and is collected at node i and delivered at node j by means of local tours. Normally $w_{ij} \neq w_{ji}$. A demand w_{ij} which is serviced by the same local tour is not served directly, but via a hub, as in common postal operations. Let $O_i = \sum_{j \in N} w_{ij}$ and $D_i = \sum_{j \in N} w_{ji}$ be the total of demand that is originated from and destined to client $i \in N$, respectively.

Let also A be the set of possible arcs (u, v) connecting clients $u, v \in N$, $u \neq v$, that can form the local tours. Through out the work, we will use indexes $i, j, u, v, t \in N$ interchangeably to represent client nodes, and $k, m \in N$ for the hubs. Normally i and j are utilized for the hub-and-spoke (HS) part, while u, v and t are used for the local tour portion of the formulation. Further, let V be the set of available vehicles, λ_{uv} the traveling time of arc $(u, v) \in A$ including the loading and unloading times at node v , and T the maximum time allowed for the tours.

The cost parameters are: a_k , the fixed cost of installing a hub at node $k \in N$; $\hat{c}_{ik}$, the cost of handling the incoming and outgoing demands of client $i \in N$ by hub $k \in N$, normally these cost are proportional to O_i and D_i ; $\check{c}_{uv}$, the cost of traveling arc $(u, v) \in A$ by a vehicle; $\hat{c}_i$, the cost of assigning a vehicle l to a hub; and $\check{c}_{ij}^{km}$, the transportation cost of demands w_{ij} and w_{ji} for the inter-hub connection (k, m) ($k, m \in N$). Remark that as the nodes are single allocated to the hubs, all the incoming and outgoing flow of a node uses the same hub, thus $\check{c}_{ij}^{km} = (w_{ij}c_{km} + w_{ji}c_{mk})\alpha$, where c_{km} is the unit transportation cost of inter-hub connection (k, m) and $0 < \alpha < 1$ represents the economies of scale.

The HS part of the formulation uses the following variables. Flow variables $x_{ijkm} \geq 0$ to represent the fraction of the demands w_{ij} and w_{ji} ($i, j \in N : i < j$) that is routed through hubs k and m ($k, m \in N$), in this order. When only one hub is used, then x_{ijkk} is set to one and no inter-hub cost is incurred. Integer variables $z_{ik} \in \{0, 1\}$ to indicate if node $i \in N$ is either allocated to hub $k \in N$ ($z_{ik} = 1$) or not ($z_{ik} = 0$). When a hub is located at node $k \in N$, then $z_{kk} = 1$; otherwise $z_{kk} = 0$.

The local tour portion of the formulation utilizes the following variables. Integer variables $q_{kl} \in \{0, 1\}$ to denote if vehicle $l \in V$ is assigned to hub $k \in N$ ($q_{kl} = 1$) or not ($q_{kl} = 0$); $y_{uv}^{kl} \in \{0, 1\}$ to represent if vehicle $l \in V$ of hub $k \in N$ uses arc $(u, v) \in A$ in its tour ($y_{uv}^{kl} = 1$) or not ($y_{uv}^{kl} = 0$); $p_v^{kl} \in \{0, 1\}$ to express if node $v \in N$ is serviced by vehicle $l \in V$ of hub $k \in N$ ($p_v^{kl} = 1$) or not ($p_v^{kl} = 0$). In order to eliminate sub-tours, auxiliary multi-commodity flow variables $f_{uv}^{klt} \geq 0$ are used. Variables f_{uv}^{klt} represent if vehicle $l \in V$ of hub $k \in N$ uses arc $(u, v) \in A$ in its tour to service node $t \in N$. The idea behind variables f_{uv}^{klt} is if a node $t \in N$ is allocated to a hub $k \in N$, then a unit of a fictitious demand must leave hub $k \in N$ and arrive at node $t \in N$ by means of a path. With the aforementioned definitions, the MMHLRP can be stated as follows:

$$\min \sum_{k \in N} a_k z_{kk} + \sum_{i \in N} \sum_{k \in N} \hat{c}_{ik} z_{ik} + \sum_{l \in V} \sum_{(u,v) \in A} \sum_{k \in N} \check{c}_{uv} y_{uv}^{kl} + \sum_{l \in V} \sum_{k \in N} \hat{c}_l q_{kl} + \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} \check{c}_{ij}^{km} x_{ijkm} \quad (1)$$

$$\text{s.t. : } \sum_{k \in N} z_{ik} = 1 \quad \forall i \in N \quad (2)$$

$$z_{ik} \leq z_{kk} \quad \forall i, k \in N : i \neq k \quad (3)$$

$$\sum_{m \in N} x_{ij}^{km} = z_{ik} \quad \forall i, j, k \in N : i < j \quad (4)$$

$$\sum_{k \in N} x_{ij}^{km} = z_{jm} \quad \forall i, j, m \in N : i < j \quad (5)$$

$$\sum_{(u,v) \in A} y_{uv}^{kl} = p_u^{kl} \quad \forall u, k \in N, l \in V \quad (6)$$

$$\sum_{(u,v) \in A} y_{uv}^{kl} = p_v^{kl} \quad \forall v, k \in N, l \in V \quad (7)$$

$$y_{uv}^{kl} \leq q_{kl} \quad \forall k \in N, (u, v) \in A, l \in V \quad (8)$$

$$q_{kl} \leq q_{k(l-1)} \quad \forall k \in N, l \in V : l > 1 \quad (9)$$

$$\sum_{l \in V} p_t^{kl} = z_{tk} \quad \forall t, k \in N : k \neq t \quad (10)$$

$$\sum_{(u,v) \in A} \lambda_{uv} y_{uv}^{kl} \leq T \quad \forall k \in N, l \in V \quad (11)$$

$$f_{uv}^{klt} \leq y_{uv}^{kl} \quad \forall (u, v) \in A, k, t \in N, l \in V : k \neq t, k \neq v, u \neq t \quad (12)$$

$$\sum_{(k,v) \in A} f_{kv}^{klt} = p_t^{kl} \quad \forall k, t \in N, l \in V : k \neq t \quad (13)$$

$$\sum_{(u,t) \in A} f_{ut}^{klt} = p_t^{kl} \quad \forall k, t \in N, l \in V : k \neq t \quad (14)$$

$$\sum_{(u,v) \in A} f_{uv}^{klt} = \sum_{(v,u) \in A} f_{vu}^{klt} \quad \forall v, k, t \in N, l \in V : k \neq t, v \neq k, v \neq t \quad (15)$$

$$x_{ij}^{km} \geq 0 \quad \forall i, j, k, m \in N : i < j \quad (16)$$

$$f_{uv}^{klt} \geq 0 \quad \forall (u, v) \in A, t, k \in N, l \in V : v \neq k, u \neq t, k \neq t \quad (17)$$

$$z_{ik} \in \{0, 1\} \quad \forall i, k \in N \quad (18)$$

$$q_{kl} \in \{0, 1\} \quad \forall k \in N, l \in V \quad (19)$$

$$p_t^{kl} \in \{0, 1\} \quad \forall k, t \in N, l \in V : k \neq t \quad (20)$$

$$y_{uv}^{kl} \in \{0, 1\} \quad \forall (u, v) \in A, k \in N, l \in V \quad (21)$$

The objective function (1) minimizes the sum of the installation, the handling, the execution of the local tours, the allocation of vehicles to the hubs, and the inter-hub transportation costs. While constraints (2)–(5) are responsible for installing hubs and allocating non-hub nodes to hubs, constraints (6)–(15) form the local tours. Constraints (2) assure that all nodes are allocated to a hub. Constraints (3) allow a node to be allocated to a hub only if that hub is installed. Constraints (4) guarantee that the exchange of flow between the origin–destination pair i and j , with node $i \in N$ being serviced by hub $k \in N$, will only exist if node $i \in N$ is first allocated to hub $k \in N$. Likewise, constraints (5) impose the same logic to node $j \in N$ and hub $m \in N$. Constraints (6) and (7) establish that an arc must leave and arrive, respectively, at a node, if a vehicle of a hub is assigned to service that node. Constraints (8) are activation constraints. A vehicle $l \in V$ of hub $k \in N$ can only use arc $(u, v) \in A$ if this vehicle is first allocated to the hub. Constraints (9) are responsible for sorting the activation of vehicles at the hubs. Constraints (10) assure that if a node $t \in N$ is allocated to hub $k \in N$, then a vehicle $l \in V$ must service it. Constraints (11) guarantee that the maximum time allowed for each tour is respected. Constraints (12)–(15) are path inducing constraints for each vehicle $l \in V$ that is set to a hub $k \in N$ and service a node $t \in N$. Finally, constraints (16)–(19), (refr17b), (21) are the standard non-negativity and integrality constraints.

Formulation (2)–(21) has one interesting feature: for fixed values of the complicating variables (integer variables), the resulting problem can be split into two smaller, simpler subproblems. While one is a transportation problem involving only the x variables, the other is a feasibility problem, i.e. it assesses if the fixed values for the y, q and p variables form valid tours. Furthermore, each subproblem can be further decomposed: one subproblem for each pair i and $j, i, j \in N, i \neq j$, for the former; and one subproblem for each $k - l - t, k, t \in N$ and $l \in V, k \neq t$, for the latter. This feature can be efficiently exploited by decomposition methods, such as the Benders decomposition method [34].

3. Benders decomposition algorithm

Benders [34] proposed a decomposition method for solving mixed integer programming (MIP) problems that have the constraint set with a special structure: when fixing the complication variables (integer variables), the mathematical program reduces to a simple, easy to solve linear problem. Benders decomposition based algorithms have been successfully applied to other types of hub-and-spoke problems [38–44], as well as other applications [45–48]. The technique relies on projection, followed by outer linearization, and using solution strategies of dualization, and relaxation [49,50].

Generally speaking, the continuous variables induce the large scale of the original model. After projecting these variables out, an equivalent model with fewer variables – usually the ones restricted to be integer in the original problem – but possibly many more constraints is obtained. By observing that a large number of these constraints will not be binding at an optimal solution, a strategy of relaxation, in which all but a few of these constraints are ignored, can be deployed. These constraints are then generated on demand by iteratively solving two levels of coordination. At a higher level, a relaxed version of the original problem with the set of the integer variables and its respective constraints is responsible for fixing the values of these integer variables and for providing a lower bound (LB) for the problem. This level is known as the master problem (MP). At a lower level, the dual of the original problem with the values of the integer variables temporarily fixed by the MP is used to produce a Benders cut to be added to the MP and provides an upper bound (UB) for the problem. This stage is known as the subproblem (SP). The algorithm alternates between the solution of MP and the SP until the UB and the LB converge towards an optimal solution, if one exists.

3.1. Benders subproblems for the MMHLRP

Let $\mathcal{Q} = \{(z, y, p, q) : (2), (3), (6)–(11) \text{ and } (18)–(21) \text{ hold}\}$ be the set of feasible values for the variables z, y, p , and q . Then for fixed values $z = \bar{z}, y = \bar{y}, p = \bar{p}$, and $q = \bar{q}$ such that $(\bar{z}, \bar{y}, \bar{p}, \bar{q}) \in \mathcal{Q}$, i.e. after installing the hubs, allocating the non-hub nodes to the hubs, assigning vehicles to the hubs and selecting arcs which might form valid tours, a primal linear SP is obtained:

$$\min \sum_{i \in N} \sum_{j \in N} \sum_{k \in N} \sum_{m \in N} \check{c}_{ij}^{km} x_{ij}^{km} \quad (22)$$

$$\text{s.t. : } \sum_{m \in N} x_{ij}^{km} = \bar{z}_{ik} \quad \forall i, j, k \in N : i < j \quad (23)$$

$$\sum_{k \in N} x_{ij}^{km} = \bar{z}_{jm} \quad \forall i, j, m \in N : i < j \quad (24)$$

$$x_{ij}^{km} \geq 0 \quad \forall i, j, k, m \in N : i < j \quad (25)$$

$$f_{uv}^{klt} \leq \bar{y}_{uv}^{kl} \quad \forall (u, v) \in A, k, t \in N, l \in V : k \neq t, k \neq v, u \neq t \quad (26)$$

$$\sum_{(k, v) \in A} f_{kv}^{klt} = \bar{p}_t^{kl} \quad \forall k, t \in N, l \in V : k \neq t \quad (27)$$

$$\sum_{(u, t) \in A} f_{ut}^{klt} = \bar{p}_t^{kl} \quad \forall k, t \in N, l \in V : k \neq t \quad (28)$$

$$\sum_{(u, v) \in A} f_{uv}^{klt} = \sum_{(v, u) \in A} f_{vu}^{klt} \quad \forall v, k, t \in N, l \in V : k \neq t, v \neq k, v \neq t \quad (29)$$

$$f_{uv}^{klt} \geq 0 \quad \forall (u, v) \in A, t, k \in N, l \in V : v \neq k, u \neq t, k \neq t \quad (30)$$

Observing SP (22)–(30), it is trivial to see that (22)–(30) are independent of one another and thus separable, being actually two SPs. Eqs. (22)–(25) form a transportation problem, while (26)–(30) are a feasibility subsystem, which verifies if the selected arcs form valid tours.

3.1.1. The transportation subproblem

Inspecting the transportation subproblem (22)–(25) closely, one can see that it can be further decomposed into simpler subsystems, one for each pair i and j ($i, j \in N : i < j$), stated as:

$$\min \sum_{k \in N} \sum_{m \in N} \check{c}_{ij}^{km} x_{ij}^{km} \quad (31)$$

$$\text{s.t. : } \sum_{m \in N} x_{ij}^{km} = \bar{z}_{ik} \quad \forall k \in N \quad (32)$$

$$\sum_{k \in N} x_{ij}^{km} = \bar{z}_{jm} \quad \forall m \in N \quad (33)$$

$$x_{ij}^{km} \geq 0 \quad \forall k, m \in N \quad (34)$$

Constraints (32) and (33) assure that an inter-hub connection $(k, m), k, m \in N$, is chosen when nodes $i, j \in N$ are allocated to hubs k and $m, k, m \in N$, respectively, i.e. when $\bar{z}_{ik} = 1$ and $\bar{z}_{jm} = 1$. Further, let $\gamma_{ijk} \in \mathbb{R}$ and $\delta_{ijm} \in \mathbb{R}$ be the dual variables associated with constraints (32) and (33), respectively. Then, for each pair i and j ($i, j \in N : i < j$), the dual subproblem of the transportation SP can be stated as follows:

$$\max \sum_{k \in N} \bar{z}_{ik} \gamma_{ijk} + \sum_{m \in N} \bar{z}_{jm} \delta_{ijm} \quad (35)$$

$$\text{s.t. : } \gamma_{ijk} + \delta_{ijm} \leq \check{c}_{ij}^{km} \quad \forall k, m \in N \quad (36)$$

$$\gamma_{ijk} \in \mathbb{R} \quad \forall k \in N \quad (37)$$

$$\delta_{ijm} \in \mathbb{R} \quad \forall m \in N \quad (38)$$

Observe that the feasible set of solutions given by (36)–(38) is not modified when the values for $\bar{z}$ vary. Hence, by strong duality, either the primal transportation SP (31)–(34) is feasible and bounded, or it is infeasible. Nevertheless it is possible to establish, following the results of Proposition 1, that constraints (2) and (3), such that $0 \leq z_{ik} \leq 1, \forall i, k \in N$, are sufficient to produce bounded subproblems.

Proposition 1. For any vector $z \in \mathcal{Z}$, such that $\mathcal{Z} = \{z : (2), (3), 0 \leq z_{ik} \leq 1, \forall i, k \in N\}$, the primal (31)–(34) and dual (35)–(38) subproblems are feasible and bounded.

Proof. For any vector $z \in \mathcal{Z}$, a unique feasible optimal solution for the primal subproblem is readily available which is $x_{ijkm} = 1$ whenever there exists at least one possible inter-hub connection k and m (x_{ijkm}) that is possible to flow the demand of the pair of nodes i and j , since i will be partially allocated to some hub k , while j will be partially assigned to some hub m . Thus the primal subproblem (31)–(34) is always feasible. Further, since the transportation costs $\check{c}_{ij}^{km}$ are non-negative and finite, and the primal feasible space (32)–(34) is bounded, then any feasible solution of the primal subproblem is always bounded. Then by strong duality, the dual subproblem (35)–(38) is also feasible and bounded. $\square$


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for  $i, j \in N : i \neq j$  do
   $\gamma_{ijs(i)} \leftarrow 0$ 
   $\delta_{ijs(j)} \leftarrow \tilde{c}_{ij}^{s(i)s(j)}$ 
  for  $m \in N : m \neq s(j)$  do
     $\delta_{ijm} \leftarrow 0$ 
  end for
  for  $k \in N : k \neq s(i)$  do
     $\gamma_{ijk} \leftarrow \max \{0; \tilde{c}_{ij}^{ks(j)} - \tilde{c}_{ij}^{s(i)s(j)}\}$ 
  end for
end for

```

Fig. 2. Explicit solution of the dual subproblem given an integer feasible hub assignment.

Furthermore, whenever a integral z vector is at hand, a unique feasible solution, obviously an optimal for the primal subproblem, is readily available. This solution consists in assigning $x_{ijkm} = 1$ to the unique pair of hubs such that $z_{ik} = 1$ and $z_{jm} = 1$, which, by the complementary slackness Theorem, allows us to explicitly solve the dual subproblem (35)–(38). Fig. 2 presents the computation of the optimal values for the dual variables, where $s(i) = k$ is a function which returns the hub ($k \in N$) assignment for a given node ($i \in N$).

Based on Proposition 1, it is possible to define the set $\mathcal{D}_{ij}$ of feasible solutions of the dual the subproblem (35)–(38) and the set $\mathcal{P}_{\mathcal{D}_{ij}}$ of extreme points of $\mathcal{D}_{ij}$, allowing then to rewrite the dual SP (35)–(38) as:

$$\max_{(\gamma, \delta) \in \mathcal{P}_{\mathcal{D}_{ij}}} \sum_{k \in N} \tilde{z}_{ik} \gamma_{ijk} + \sum_{m \in N} \tilde{z}_{jm} \delta_{ijm}$$

With the help of an additional variable $\eta_{ij} \geq 0$, representing an inter-hub cost estimator for the demand of the pair of nodes i and j ($i, j \in N : i < j$), one can produce the following optimality Benders cuts:

$$\eta_{ij} \geq \sum_{k \in N} \tilde{\gamma}_{ijk} \tilde{z}_{ik} + \sum_{m \in N} \tilde{\delta}_{ijm} \tilde{z}_{jm} \quad \forall (\tilde{\gamma}, \tilde{\delta}) \in \mathcal{P}_{\mathcal{D}_{ij}}, i, j \in N : i < j \quad (39)$$

These optimality Benders cuts (39) will be used to formulate the Benders master problem in Section 3.2.

3.1.2. The feasibility subproblem

Now, recalling the feasibility subproblem (26)–(30), it is also trivial to see that it can be separated into smaller subsystems, one for each triple k, l, t ($k, t \in N, l \in V : k \neq t$):

$$\min \mathbf{0} \quad (40)$$

$$\text{s.t. : } f_{uv}^{klt} \leq \bar{y}_{uv}^{kl} \quad \forall (u, v) \in A : k \neq v, u \neq t \quad (41)$$

$$\sum_{(k, v) \in A} f_{kv}^{klt} = \bar{p}_t^{kl} \quad (42)$$

$$\sum_{(u, t) \in A} f_{ut}^{klt} = \bar{p}_t^{kl} \quad (43)$$

$$\sum_{(u, v) \in A} f_{uv}^{klt} = \sum_{(v, u) \in A} f_{vu}^{klt} \quad \forall v \in N : v \neq k, v \neq t \quad (44)$$

$$f_{uv}^{klt} \geq 0 \quad \forall (u, v) \in A : v \neq k, u \neq t \quad (45)$$

where (40) is a dummy objective function. Subsystem (40)–(45) verifies if the assigned vehicle to the hub to service the non-hub nodes allocated to the hub, represented by $\bar{p}$, and the selected arcs, represented by $\bar{y}$, form a valid solution or not. In other words, it verifies if the values $(\bar{p}, \bar{q}, \bar{y})$ are a feasible solution for the subproblem (40)–(45) or not.

Let $\delta_{uv}^{klt} \geq 0$, $\mu_{klt} \in \mathbb{R}$, $\beta_{klt} \in \mathbb{R}$, and $\psi_v^{klt} \in \mathbb{R}$ be the dual variables associated with constraints (41)–(44), respectively. The dual subproblem of the feasibility subsystem can be given, for each triple k, l , and t ($k, t \in N, l \in V : k \neq t$), as:

$$\max \bar{p}_{klt} \beta_{klt} + \bar{p}_{klt} \mu_{klt} - \sum_{\substack{(u, v) \in A \\ u \neq t, v \neq k}} \bar{y}_{uv}^{kl} \delta_{uv}^{klt} \quad (46)$$

$$\text{s.t. : } \psi_v^{klt} - \psi_u^{klt} - \delta_{uv}^{klt} \leq 0 \quad \forall (u, v) \in A : u \neq t, v \neq k \quad (47)$$

$$\mu_{klt} + \psi_v^{klt} - \delta_{kv}^{klt} \leq 0 \quad \forall (k, v) \in A : v \neq t \quad (48)$$

$$\beta_{klt} - \psi_u^{klt} - \delta_{ut}^{klt} \leq 0 \quad \forall (u, t) \in A : u \neq k \quad (49)$$

$$\mu_{klt} + \beta_{klt} - \delta_{kt}^{klt} \leq 0 \quad (50)$$

$$\delta_{uv}^{klt} \geq 0 \quad \forall (u, v) \in A \quad (51)$$

$$\mu_{klt} \in \mathbb{R} \quad (52)$$

$$\beta_{klt} \in \mathbb{R} \quad (53)$$

$$\psi_v^{klt} \in \mathbb{R} \quad \forall v \in N \quad (54)$$

When the primal subproblem (40)–(45) is infeasible then, by strong duality, its dual (46)–(54) is infeasible or unbounded. However, as the null vector is a dual feasible solution, the problem (46)–(54) is unbounded whenever the system (40)–(45) is infeasible. If the original model is feasible, then the cone $\mathbb{C}_{klt}$, formed by constraints (47)–(54), is a pointed cone, having the origin as a vertex. As any solution in $\mathbb{C}_{klt}$ is a ray, and as $\mathbb{C}_{klt}$ has a finite number of generators, it is then possible to rewrite the subproblem (46,47) as:

$$\max_{(\bar{\delta}, \bar{\mu}, \bar{\beta}, \bar{\psi}) \in \mathbb{C}_{klt}} \bar{p}_{klt} \bar{\beta}_{klt} + \bar{\mu}_{klt} \bar{\mu}_{klt} - \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \bar{y}_{uv}^{kt} \bar{\delta}_{klt}^{uv}$$

which is bounded if and only if

$$\bar{\beta}_{klt} \bar{p}_{klt} + \bar{\mu}_{klt} \bar{\mu}_{klt} - \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \bar{\delta}_{klt}^{uv} \bar{y}_{uv}^{kt} \leq 0 \quad \forall (\bar{\delta}, \bar{\mu}, \bar{\beta}, \bar{\psi}) \in \mathbb{C}_{klt}, k, t \in N, l \in V : k \neq t$$

By the Minkowski's theorem, every ray in $\mathbb{C}_{klt}$ can be expressed by a nonnegative linear combination of extreme rays, represented by the set $\mathcal{R}_{\mathbb{C}_{klt}}$. Hence it is possible to restrict the values of $(\delta, \mu, \beta, \psi)$ to the set of extreme rays and yet get a characterization of the feasibility of (p, q, y) , resulting then in what it is called as Benders feasibility cuts [49,54]:

$$\bar{\beta}_{klt} \bar{p}_{klt} + \bar{\mu}_{klt} \bar{\mu}_{klt} - \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \bar{\delta}_{klt}^{uv} \bar{y}_{uv}^{kt} \leq 0 \quad \forall (\bar{\delta}, \bar{\mu}, \bar{\beta}, \bar{\psi}) \in \mathcal{R}_{\mathbb{C}}, k, t \in N, l \in V : k \neq t \quad (55)$$

Using constraints (39) and (55), it is now possible to formulate the Benders master problem (MP).

3.2. Benders master problem

The Benders MP is given as:

$$\sum_{i \in N} \sum_{\substack{k \in N \\ k \neq i}} \hat{c}_{ik} z_{ik} + \sum_{l \in V} \sum_{(u,v) \in A} \sum_{k \in N} \check{c}_{uv} y_{uv}^{kl} + \sum_{k \in N} a_k z_{kk} + \sum_{l \in V} \sum_{k \in N} \hat{c}_l q_{kl} + \sum_{i \in N} \sum_{\substack{j \in N \\ i < j}} \eta_{ij} \quad (56)$$

s.t.: (2), (3), (6)–(11) and (18)–(21)

$$\eta_{ij} \geq \sum_{k \in N} \tilde{\gamma}_{ijk} z_{ik} + \sum_{m \in N} \tilde{\delta}_{ijm} z_{jm} \quad \forall (\tilde{\gamma}, \tilde{\delta}) \in \mathcal{P}_{Dij}, \quad i, j \in N : i < j \quad (57)$$

$$\bar{\beta}_{klt} \bar{p}_{klt} + \bar{\mu}_{klt} \bar{\mu}_{klt} - \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \bar{\delta}_{klt}^{uv} \bar{y}_{uv}^{kt} \leq 0 \quad \forall (\bar{\delta}, \bar{\mu}, \bar{\beta}, \bar{\psi}) \in \mathcal{R}_{\mathbb{C}_{klt}}, k, t \in N, l \in V : k \neq t \quad (58)$$

$$\eta_{ij} \geq 0 \quad \forall i, j \in N : i < j \quad (59)$$

The Benders MP (56)–(59) is equivalent to the original problem (2)–(21), but it has fewer variables, and probably an exponential number of constraints. Fortunately, in an optimal solution, just some of these constraints will be active. It is then possible to tackle the Benders MP (56)–(59) by means of a strategy of relaxation, where all but a few of the constraints (57) and (58) are disregarded. This requires an adequate cutting plane procedure where relaxed MPs, having a restricted subset of the constraints (57) and (58), are solved iteratively. This subset is enriched by new cuts associated with extreme points of $\mathcal{P}_D$ and extreme rays of $\mathcal{R}_{\mathbb{C}}$ when needed. In order to generate these cuts, the respective dual SPs are solved at each iteration. The procedure goes on until an optimal solution is obtained.

As the Benders MP is now a relaxed formulation of the original problem, it provides therefore a lower bound (LB) for it. This LB is improved by the addition of Benders cuts at each iteration, whenever the feasibility dual SP is bounded. When the LB converges to an upper bound (UB), an optimal solution of the original problem is attained, if one exists.

In order to compute an UB, whenever the optimal solution value $\phi_{FSP}(\delta^h, \mu^h, \beta^h, \psi^h)$ of the feasibility dual SP is bounded, one can use the optimal solution value $\phi_{TSP}(\gamma^h, \delta^h)$ of the transportation SP added to the value of $\phi_{INT}(z^h, p^h, q^h, y^h)$, where δ^h, μ^h, β^h , and ψ^h , and γ^h and δ^h are the optimal values for the dual variables of the feasibility and the transportation dual SPs, respectively; and z^h, p^h, q^h , and y^h are optimal values for the integer variables of the MP for iteration $h \in \{1, \dots, H\}$, and H is the maximum number of iterations to attain optimality. Further, $\phi_{INT}(z^h, p^h, q^h, y^h)$ is defined as $\sum_{i \in N} \sum_{k \in N} \hat{c}_{ik} z_{ik}^h + \sum_{l \in V} \sum_{(u,v) \in A} \sum_{k \in N} \check{c}_{uv} y_{uv}^{klh} + \sum_{k \in N} a_k z_{kk}^h + \sum_{l \in V} \sum_{k \in N} \hat{c}_l q_{kl}^h$. An outline of a basic Benders decomposition algorithm is presented in Fig. 3: where $\phi_{MP}(\eta^h, z^h, p^h, q^h, y^h)$ is the optimal value for the objective function of the MP of iteration h .

The speed of convergence of the Benders decomposition algorithm relies substantially on the strength of the Benders cuts generated, i.e. on the selection of the values for the dual variables. Strong cuts usually result in fewer iterations. As the dual SPs of a given iteration are usually degenerate, having thus alternative optimal solutions, it is possible to deploy tailored schemes for selecting appropriate values for the dual variables. These schemes help to greatly improve the overall performance of the method.

3.3. Tailoring the Benders decomposition algorithm

Since there are two different types of dual subproblems, two different types of cut selection schemes are used to improve the overall performance of the method: one for the transportation subproblems (35)–(38), and another for the feasibility subsystems (46)–(54).

3.3.1. Pareto-optimal cuts

The convergence of the Benders decomposition algorithm is improved when non-dominated cuts, known as Pareto-optimal cuts, are applied. Magnanti and Wong [51] were the first authors to present a procedure capable of generating these cuts. They noticed that when the dual SPs have multiple optimal solutions different Benders cuts can be produced. Instead of adding them all to the MP, they proposed a slightly different dual SP in order to find the strongest non-dominated cut in the sense of Pareto-optimality. In the present context, the strength of a cut according to [51] can be stated as the following definition:

Definition 1. A cut is Pareto-optimal if it is not dominated by any other cut. A Benders cut (39) corresponding to $(\gamma^1, \delta^1) \in \mathcal{D}_{ij}$ dominates the cut corresponding to $(\gamma^2, \delta^2) \in \mathcal{D}_{ij}$, if:

$$\sum_{k \in N} \gamma_{ijk}^1 z_{ik} + \sum_{m \in N} \delta_{ijm}^1 z_{jm} \geq \sum_{k \in N} \gamma_{ijk}^2 z_{ik} + \sum_{m \in N} \delta_{ijm}^2 z_{jm} \quad \forall z \in \mathcal{Z}, i, j \in N : i \neq j$$

with strict inequality for at least one point $z \in \mathcal{Z}$. Similarly, it is said that (γ^1, δ^1) dominates (γ^2, δ^2) and it is called Pareto-optimal.

Recall that $\mathcal{Z} = \{z : (2), (3), 0 \leq z_{ik} \leq 1, \forall i, k \in N\}$ is the set of feasible values for the variables z . Further, let $ri(\mathcal{Z})$ denote the relative interior of $\mathcal{Z}$ and $z^0 \in ri(\mathcal{Z})$ be a core point. In order to find Pareto-optimal cuts, [51] proposed the solution of two different SPs at each iteration h of the Benders algorithm: One dual SP associated with the current MP solution and another related to an initial fixed given core point z^0 . For the transportation subproblem of the MMHLRP, for each pair i and j ($i, j \in N : i < j$) the SP of [51] can be stated as:

$$\max \sum_{k \in N} z_{ik}^0 \gamma_{ijk} + \sum_{m \in N} z_{jm}^0 \delta_{ijm} \quad (60)$$

s.t. : (36)–(38)

$$\sum_{k \in N} z_{ik}^h \gamma_{ijk} + \sum_{m \in N} z_{jm}^h \delta_{ijm} = \phi_{TSP_{ij}}(\gamma^h, \delta^h) \quad (61)$$

where $\phi_{TSP_{ij}}(\gamma^h, \delta^h)$ is the optimal solution of the dual subproblem for pair i and j in iteration h .

The [51] SP (60), (61) generates Pareto-optimal cuts that help to accelerate the Benders decomposition algorithm. However, due to the dense constraint (61), the [51] SP is prone to numerical instability and requires a significant computational effort to be solved. Fortunately, [52] demonstrates that constraints (61) can be disregarded if a different core point $z^0 \in \mathcal{Z}$ is used on the objective function (60) at each iteration. Furthermore, he shows that z^0 does not even need to be a core point to produce a Pareto-optimal cut, but it can be a point of [51], proposing then the following definition.

Definition 2. If a point z^0 is not one of the points that spans the set of all possible projections into the variables of the master problem; and when the space of feasible solutions of the dual and primal subproblems are not empty, then z^0 is a point of [51].

```

UB ← ∞, LB ← 0, h ← 0
while UB ≠ LB do
  Solve MP (56)–(59) to get  $(\eta^h, z^h, p^h, q^h, y^h)$ 
  Solve dual transportation SP (35)–(38) to get  $(\gamma^h, \delta^h)$ 
  Add cut (39) to MP
  Solve feasibility SP (46)–(54)
  if  $\phi_{FSP}(\delta^h, \mu^h, \beta^h, \psi^h) < \infty$  then
    LB ←  $\phi_{MP}(\eta^h, z^h, p^h, q^h, y^h)$ 
    UB ←  $\min(UB, \phi_{TSP}(\gamma^h, \delta^h) + \phi_{INT}(z^h, p^h, q^h, y^h))$ 
  else
    Get an extreme ray  $(\delta^h, \mu^h, \beta^h, \psi^h)$ 
    Add cut (55) to MP
  end if
  h ← h + 1
end while

```

Fig. 3. A basic Benders decomposition algorithm for the MMHLRP.

These enhancements allow the dual SP of [51] to be lighter, requiring thus less computational effort to generate Pareto-optimal cuts. The drawback of [52]' improvements is the necessity of a different core point or point of [51] at each iteration. Although [53] point out that there are not practical methods for finding good core points, [52] proves that given a valid initial core point $z^0 \in \text{ri}(\mathcal{Z})$ and $z \in \mathcal{Z}$ then any convex combination of z^0 and z is also a point of [51]. This approximation scheme can be represented by:

$$z_{ik}^0 = \theta z_{ik}^0 + (1 - \theta) z_{ik}^h \quad \forall i, k, h \quad (62)$$

where $0 < \theta < 1$, but $\theta = 1/2$ gives empirically the best results, as observed also by [52,53].

For the MMHLRP, an initial point of [51] can be given as:

$$z_{kk}^0 = 0.5 \quad \forall k \quad (63)$$

$$z_{ik}^0 = 0.5/(n - 1) \quad \forall i \neq k \quad (64)$$

where $n = |N|$.

Proposition 2. The point given by (63) and (64) is a valid point of [51].

Proof. The initial value z^0 can be build by using a convex combination of $n + 1$ different feasible solution: n solutions having only a single hub, and one solution having n installed hubs. By properly selecting values for the weights of a convex combination, but observing constraints (2) and (3), it is possible to combine these $n + 1$ solutions. One possible selection of values is given by prescribing weights equal to $1/2(n - 1)$ for the first n solutions, and $(1/2 - 1/2(n - 1))$ for the last one, resulting then on the point given by (63) and (64). $\square$

3.3.2. Minimum infeasible subsystems

[54] propose a cut selection scheme specially tailored for problems having feasibility subsystems. Actually the scheme is a clever unified framework that solves the separation of both optimality and feasibility cuts yielding a single type of cut. The procedure relies on the fact that the Benders subproblem can always be reformulated as a pure feasibility problem by stating the separation of the current master solution as an additional constraint. Therefore, a violated cut is identified whenever the aforementioned system with the additional constraint is infeasible. Extreme rays can then be produced by detecting the minimal source of infeasibility or the smallest set of cuts that are capable of cutting the current MP solution.

When the primal subproblem is already a feasibility problem, one just needs to find this minimal source of infeasibility. For the subsystem (40)–(45), one way to accomplish that is by adding auxiliary variables $\mathcal{E}^+ \geq 0$ and $\mathcal{E}^- \geq 0$ on the coupling constraints involving the complicating and the large scale variables of the feasibility subsystem. The resulting primal problem, for each triple k, l, t ($k, t \in N, l \in V : k \neq t$), is then given as:

$$\begin{aligned} \min \quad & \mathcal{E}^+ + \mathcal{E}^- \\ \text{s.t.} \quad & -f_{uv}^{klt} \geq -\bar{y}_{uv}^{kl} - \mathcal{E}^- \quad \forall (u, v) \in A : k \neq v, u \neq t \\ & \sum_{(k,v) \in A} f_{kv}^{klt} = \bar{p}_t^{kl} + \mathcal{E}^+ - \mathcal{E}^- \\ & \sum_{(u,t) \in A} f_{ut}^{klt} = \bar{p}_t^{kl} + \mathcal{E}^+ - \mathcal{E}^- \\ & \sum_{(u,v) \in A} f_{uv}^{klt} = \sum_{(v,u) \in A} f_{vu}^{klt} \quad \forall v \in N : v \neq k, v \neq t \\ & f_{uv}^{klt} \geq 0 \quad \forall (u, v) \in A : v \neq k, u \neq t \end{aligned}$$

In order to generate the extreme rays for the MMHLRP, the following dual subproblem has then to be solved, one for each k, l, t ($k, t \in N, l \in V : k \neq t$):

$$\max \quad \bar{p}_{klt} \beta_{klt} + \bar{p}_{klt} \mu_{klt} - \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \bar{y}_{uv}^{kt} \delta_{klt}^{uv} \quad (65)$$

$$\begin{aligned} \text{s.t.} \quad & (47) - (54) \\ & \mu_{klt} + \beta_{klt} + \sum_{\substack{(u,v) \in A \\ u \neq t, v \neq k}} \delta_{uv}^{klt} \leq 1 \quad (66) \end{aligned}$$

$$\mu_{klt} + \beta_{klt} \geq -1 \quad (67)$$

3.3.3. The tailored Benders decomposition algorithm

The combination of the schemes of Sections 3.3.1 and 3.3.2 into a tailored Benders decomposition procedure gives raise to the algorithm shown in Fig. 4:

[48,55,42] demonstrate that embedding the Benders decomposition algorithm into a branch-and-cut frameworks has shown to be very helpful in improving the overall performance of the procedure, speeding up the method. Such strategy

is better illustrated in Fig. 5, where the pool of cuts $\mathcal{P}$ can be initialized with cuts coming from a warm start phase and adjoined by the cuts of each iteration, and by the ones generated inside the branch-and-bound tree. This branch-and-cut framework is used to solve the master problems.

4. Computational experiments

The computational experiments were carried out on an Intel Xeon with 8 cores of 2.53 GHz and 24 GB of memory, running the operational system Linux. Instances generated from the Australian Post (AP) standard data set were utilized as the test bed. This data set, introduced by [13,56], can produce instance sizes ranging from 10 up to 200 nodes. The instances are named as AP $n.\alpha$ where n is the number of nodes and $\alpha = \{2, 4, 6, 8\}$ represents the applied discount factors 0.2, 0.4, 0.6, and 0.8, respectively, on the inter-hub connections.

For sake of simplicity, the costs $\tilde{c}$ and the parameters λ were set to be equal to the Euclidean distance between the nodes involved, while the costs $\hat{c}$ were set to the Euclidean distance between the nodes involved multiplied by the sum of O_i and D_j . The costs $\tilde{c}$ and the parameter T were made equal to 1,000 and 50 units, respectively. Further, as this data set has fixed cost information for the first 50 nodes only, the fixed costs were then randomly generated for all of the nodes of all instances using a Gaussian distribution with average equal to f_0 and the coefficient of variance set to 40% in order to portray how the installation costs vary in real problems. The parameter f_0 has been introduced by [57] and represents the scaled difference in objective value between a scenario in which there is a virtual hub located in the center of mass of the demand and scenario in which all nodes are hubs.

The proposed Benders decomposition (BD) algorithm has been able to solve instances ranging from 10 up to 100 nodes. The number of integer and continuous variables of the solved problems, as well as the computer memory required to store the full diagonals of the constraint matrices, are depicted in Table 1. It is remarkable that the largest solved instance has almost 3.5 GB of data on the full diagonals, making it unmanageable on regular personal computers without the aid of decomposition techniques. Furthermore, the subproblems and the master problems are solved by using the Simplex method and the mixed integer programming solver of CPLEX, respectively.

The first set of tests compares the proposed BD algorithm with IBM CPLEX 12 for solving instances ranging from 10 to 30 nodes. Besides the running times, the number of active Benders cuts and iterations, as well as of installed hubs and the large-

```

 $UB \leftarrow \infty, LB \leftarrow 0, h \leftarrow 0, \theta \leftarrow 0.5$ 
Set  $z^0$  using (63) and (64)
while  $UB \neq LB$  do
    Solve dual transportation SP (60)–(61) to get  $(\gamma^h, \delta^h)$ 
    Add cut (39) to MP
    Solve MP (56)–(59) to get  $(\tilde{z}, \tilde{p}, \tilde{q}, \tilde{y})$ 
    Solve dual transportation SP (35)–(38) to get  $(\gamma^h, \delta^h)$ 
    Add cut (39) to MP
    Solve feasibility SP (46)–(54) to get  $(\eta^h, z^h, p^h, q^h, y^h)$ 
    if  $\phi_{FSP}(\delta^h, \mu^h, \beta^h, \psi^h) > 0$  then
        Add cut (55) to MP
    else
         $LB \leftarrow \phi_{MP}(\eta^h, z^h, p^h, q^h, y^h)$ 
         $UB \leftarrow \min(UB, \phi_{TSP}(\gamma^h, \delta^h) + \phi_{INT}(z^h, p^h, q^h, y^h))$ 
    end if
    Update  $z^0$  by using (62)
     $h \leftarrow h + 1$ 
end while

```

Fig. 4. A tailored Benders decomposition algorithm for the MMHLRP.

```

Set branch-and-bound tree  $\mathcal{T} = \{o\}$ , where  $o$  has no branching constraints, but has the cuts of
pool  $\mathcal{P}$ 
while  $\mathcal{T}$  is nonempty do
    Select a node  $o' \in \mathcal{T}$ 
     $\mathcal{T} \leftarrow \mathcal{T} \setminus \{o'\}$ 
    Solve  $o'$  and let  $(\tilde{z}, \tilde{p}, \tilde{q}, \tilde{y})$  be an optimal solution
    if  $(\tilde{z}, \tilde{p}, \tilde{q}, \tilde{y})$  is a new feasible solution then
        Add Benders cuts to  $\mathcal{P}$ 
         $\mathcal{T} \leftarrow \mathcal{T} \cup \{o'\}$ 
    else
        Branch, resulting in nodes  $o''$  and  $o'''$ 
         $\mathcal{T} \leftarrow \mathcal{T} \cup \{o'', o'''\}$ 
    end if
end while

```

Fig. 5. A branch-and-cut framework for the Benders algorithm.

Table 1

Formulation dimensions for the solved instances.

# Nodes	# Integer variables	# Continuous variables	Full diagonal size (bytes)
10	4,140	30,330	275,760
20	32,480	593,560	5,008,320
30	109,020	3,207,690	26,533,680
40	257,760	10,470,700	85,827,680
50	502,700	26,060,600	212,506,400
60	867,840	54,735,500	444,826,720
70	1,377,180	102,333,000	829,681,440
80	2,054,720	175,772,000	1,422,613,760
90	2,924,460	283,049,000	2,287,787,680
100	4,010,400	433,244,000	3,498,035,200

Table 2

Comparing the BD algorithm with CPLEX 12.

Instance	BD			# Hubs	# Max. vehicles	CPLEX Time [s]	Speed-up CPLEX/BD
	# Cuts	# Iterations	Time [s]				
Ap10.2	46	3	3.02	3	2	22.96	7.60
Ap10.4	17	3	4.85	2	2	29.49	6.08
Ap10.6	26	4	19.41	1	4	40.1	2.07
Ap10.8	43	3	7.87	1	4	46.15	5.86
Ap20.2	21	4	54.3	3	2	2,419.98	44.57
Ap20.4	6	4	67.11	2	3	2,915.31	43.44
Ap20.6	17	3	732.64	2	3	7,611.47	10.39
Ap30.2	7	2	192.2	5	2	17,637.87	91.77

Table 3Results for different values of the discount factors α .

Instance	BD					# Hubs	# Max. vehicles
	# Cuts	# Iterations	GAP (LP/MILP)	# B&B nodes	Time [s]		
Ap10.2	46	3	0.0025	6	3.02	3	2
Ap10.4	17	3	0.0024	2	4.85	2	2
Ap10.6	26	4	0.0041	829	19.41	1	4
Ap10.8	43	3	0.0041	1158	7.87	1	4
Ap20.2	21	4	0.01	556	54.3	3	2
Ap20.4	6	4	0.0041	1304	67.11	2	3
Ap20.6	17	3	0.0143	2465	732.64	2	3
Ap20.8	36	3	0.0322	12,759	609.65	2	3
Ap30.2	7	2	0.0084	340	192.2	5	2
Ap30.4	47	7	0.0171	12,074	988.02	6	2
Ap30.6	41	7	0.011	1972	805.76	7	2
Ap30.8	28	6	0.0066	1235	698.05	7	2
Ap40.2	16	4	0.0109	11,226	1967.95	5	3
Ap40.4	22	3	0.0092	1777	1679.71	6	2
Ap40.6	51	7	0.0116	8779	4487.18	7	1
Ap40.8	85	13	0.0116	27,547	9408.04	10	2
Ap50.2	8	5	0.0113	8991	3969.09	6	3
Ap50.4	9	5	0.0105	24,977	5361.14	6	3
Ap50.6	19	5	0.0113	17,877	7141.58	6	3
Ap50.8	38	5	0.0103	11,031	7132.59	7	2

est number of assigned vehicles to a hub, are shown in Table 2. The BD algorithm clearly outperforms CPLEX by almost two orders of magnitude on the largest solved instance, please refer to column *Speed-up CPLEX/BD*.

The second set of tests assesses how the BD algorithm performs when varying the inter-hub discount factor α . Table 3 shows the obtained results. Instances ranging from 10 to 50 nodes were solved to optimality for all the considered discount factors, i.e. $\alpha = \{0.2, 0.4, 0.6, 0.8\}$ using again a tour limit T of 50 units. A closer look at Table 3 reveals that, for increasing values of α or, in other words, by reducing the hubs' efficiency on providing large discount factors on the inter-hub connections, the problem becomes more difficult to solve. This effect can be partially explained by the probable increase of the integrality gaps due to a more pronounced interference of the maximum time tour, parameter T , over the other costs. Since, for most of the instances solved, augmenting the value of α implies in fewer installed hubs, and more non-hub nodes being allo-

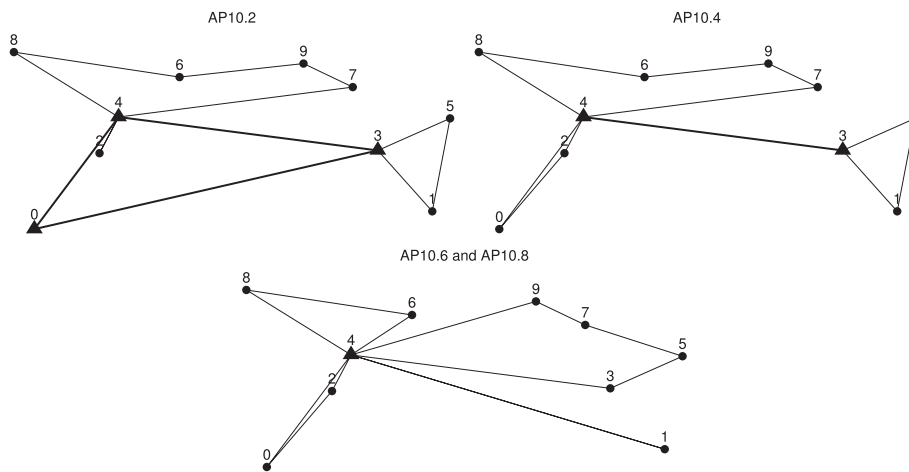
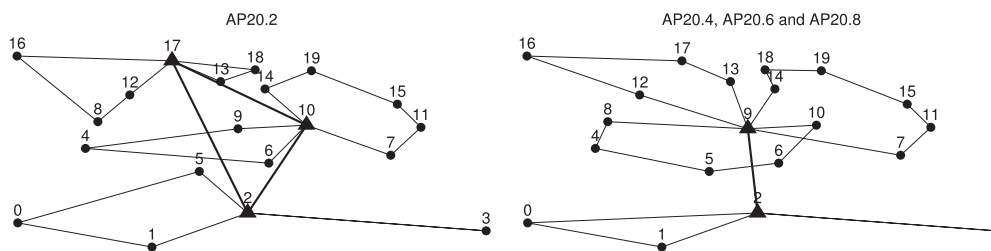
Fig. 6. Optimal network for AP10 ($T = 50$).Fig. 7. Optimal network for AP20 ($T = 50$).

Table 4
Results for different values of T for instance Ap20.4.

Parameter T	BD					# Hubs	# Max. vehicles
	# Cuts	# Iterations	GAP (LP/MILP)	# B&B nodes	Time [s]		
25	29	4	0.0734	33,944	469.6	5	4
35	60	6	0.0501	102,037	2800.2	4	4
45	50	5	0.023	14,525	477.8	3	3
50	6	4	0.0041	1304	67.11	2	3
60	5	4	0.0046	6385	97.11	2	3
70	3	6	0.0035	585	38.76	2	2
80	6	4	0.0026	116	19.35	2	2
90	10	6	0.0025	98	18.6	2	2
100	7	4	0.0025	72	14.66	2	2
150	8	3	0.0015	0	9.12	2	1

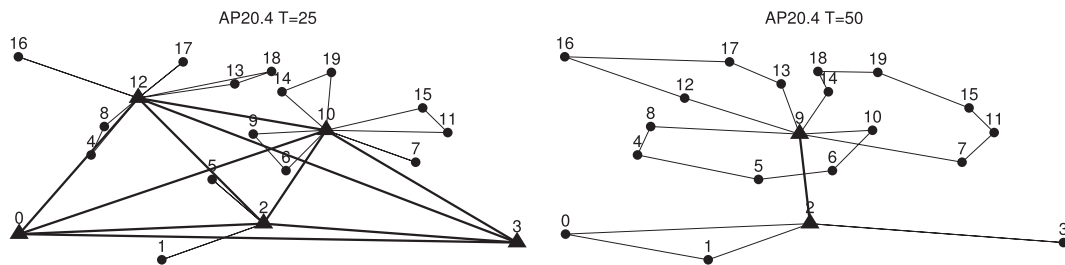
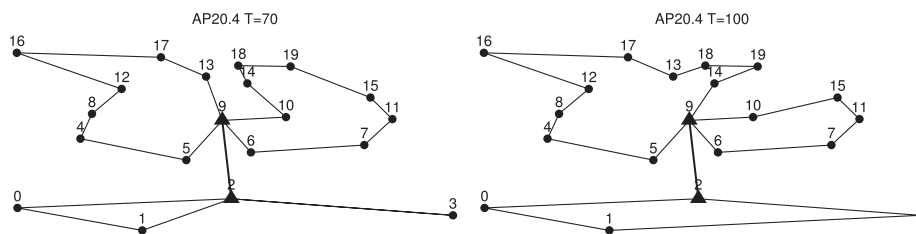
cated to a hub. Due to the tour time limit, more vehicles need to be assigned to the hubs. This results thus in a more important role of the tours. Figs. 6 and 7 illustrate the optimal obtained network topologies for instances AP10 and AP20 and provide some insights about the aforementioned effect. The optimal topologies are the same for AP10.6 and AP10.8, and for AP20.4, AP20.6, and AP20.8. As the value of α increases, the number of installed hubs tends to decrease. Nevertheless, the number of assigned vehicles per hub or tours tends to increase. Moreover, on these and on the foregoing experiments, the number of branch-and-bound nodes and the gaps of the linear programming relaxation are also reported. An interesting feature of the proposed model is that even though the linear programming gaps are very small, the number of branch-and-bound nodes required to be enumerated appears to be large as depicted on Tables 3–5. This may be explained by the large number of unfeasible nodes that have to be visited during the branch-and-bound search.

The third set of experiences is aimed to evaluate how the computational effort is affected by varying the maximum tour limit, parameter T . Instance AP20.4 was then optimally solved for different values of T . The results are shown in Table 4. As expected, the problem is hardened as the parameter T is reduced, implicating then that the knapsack constraint (11)

Table 5

From small to large scale instances.

Instance	BD					# Hubs	# Max. vehicles
	# Cuts	# Iterations	GAP (LP/MILP)	# B&B nodes	Time [s]		
Ap10.2	46	3	0.0025	6	3.02	3	2
Ap20.2	21	4	0.01	556	54.30	3	2
Ap30.2	7	2	0.0084	340	192.20	5	2
Ap40.2	16	4	0.0109	11,226	1,967.95	5	3
Ap50.2	8	5	0.0113	8,991	3,969.09	6	3
Ap60.2	9	4	0.0142	11,021	8,279.59	7	3
Ap70.2	40	9	0.0155	58,187	37,928.94	8	3
Ap80.2	18	5	0.0212	314,946	41,934.39	8	3
Ap90.2	24	7	0.0252	18,645	37,814.85	9	2
Ap100.2	40	9	0.0351	359,004	166,650.78	11	2

**Fig. 8.** Optimal networks for AP20.4 for $T = 25$ and $T = 50$.**Fig. 9.** Optimal networks for AP20.4 for different values of $T = 70$ and $T = 100$.

dominates the problem. As a consequence, the integrality gaps were enlarged. Future research may consider using *Fenchel cutting planes* to overcome these difficulties as suggested by [58]. Again, in order to provide a visual aid on how the topology is changed by altering the tour limit T , Figs. 8 and 9 depict the attained networks for values 25, 50, 70 and 100. As can be seen, for $T = 25$, hubs are installed just to be close to the non-hub nodes. As the value of T is increased, the configuration of hubs is maintained, though the number of assigned vehicles are decreased, leading to more efficient routes. Further, from Table 4, it can be inferred that the selected instances to be solved are not the most difficult ones, but they are not the easiest ones either. It can be said that the instances have a medium level of solving difficulty.

The final set of experiments tries to establish which sizes of instances are capable of being solved by the BD algorithm on the machine used. Here, the adopted discount factor for all the experiments is 0.2 and the maximum time tour limit selected is 50 units. The algorithm was able to solve instances ranging from 10 up to 100 nodes under 48 h of computing time. Table 5 shows all the obtained results. Recall that the largest solved instance has more than 4 millions of integer variables, ensuring therefore the efficiency of Benders decomposition algorithm for this important problem.

5. Conclusions

A new formulation and a tailored Benders decomposition algorithm are presented here for the many-to-many hub location routing problem. The developed algorithm relies on Pareto optimal cuts and minimum infeasible subsystems. The produced strong cuts were used in a standard *Branch-And-Cut* scheme and have shown to be very effective on solving the problem. Despite of the size of the proposed model, instances of up to 100 nodes can be solved by the method, clearly confirming the efficiency and robustness of the algorithm. To the best of the authors knowledge, there is no previous report of

such large instances of this problem being solved to optimality. Further research may address how to improve the current formulation using *Fenchel cutting planes* or how to extend it by considering hub congestion, or other routing constraints.

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